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Machine learning-based classification of the traffic of digital marketing campaigns

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***Abstract*—Digital marketing i.e., any type of advertising proposed on electronic devices over the Internet, has been the prevalent advertising methodology in the last decades. As such, measuring its effectiveness and distinguishing genuine conversions generated by real users from fraudulent traffic (for example generated by bots) is of utmost importance. In this paper, such issue is tackled by using Machine Learning and, specifically, comparing a Shallow Neural Network and a Random Forest, to automatically classify traffic data of digital marketing campaigns into "good" and "bad" traffic. The two proposed models are evaluated on real traffic data, manually annotated. The results demonstrate the feasibility of the proposed classification task, with the Shallow Neural Network achieving a 98% precision and recall, without exhibiting overfitting, over the 32,000 samples of the tested dataset.**

***Index Terms*—Digital Marketing, Bots, Web Traffic, Lead-generation, Neural Network, Random Forest**

I. INTRODUCTION

Digital marketing has become one of the most used and prevalent methods of advertising in the last couple of decades with the diffusion of the Internet and especially of social media, surpassing other methods such as television, radio and magazines [1]. Specifically, in 2023, 63% of marketers say their biggest content challenge is driving traffic and generating leads, whereas traditional marketing generates 50% fewer interactions with customers than digital marketing [2]. The process of marketing has fundamentally changed over the years, going from general advertisements that were aimed at reaching a wide audience to personalized and targeted ads and messages for every single user, a process known as "People-based marketing" [3]. For example, in 2016, almost 63% of advertisers said that they had seen improved click-through rates with addressable media and 60% had experienced higher conversion rates [4].

Digital marketing is a broad domain: any and all advertising that uses the Internet and an electronic device such as a computer, smartphone or tablet to reach its audience can be defined as digital marketing [5]. As such, digital marketing takes different shapes, such as video ads, image banners displayed on pages, suggested links during the use of search engines, emails, and sponsored posts on social media. One peculiar aspect of digital marketing campaigns is the lead generation i.e., the execution of a specific marketing strategy on some (or all) of these digital channels by a brand with the purpose of defining if a specific user is potentially interested in buying a service or a product, or to improve the brand's conversion rate [6]. Briefly, this is referred to as "to generate a lead".

Given the listed premises, to execute an efficient digital marketing campaign and measure its effectiveness, it is necessary to monitor a very large amount of data related to the internet traffic that passes through these channels, keeping track of features such as page views, clicks on links, and time spent on the page. One of the biggest problems related to the monitoring of the performance of digital marketing campaigns and such metrics is the capability to distinguish which of these data are trustworthy (real page views or clicks done by a potential customer) and which are fraudulent or automatically generated (by bots) [7]. It is important to distinguish real leads from those generated, for example, by using VPNs or TOR software, or by the same IP (as it is not trustworthy the case where the same IP generates multiple leads). In fact, according to the "2022 Imperva Bad Bot Report" [8], "bad bots traffic" surpassed human activity on the internet. Specifically, bot traffic constituted 42.3% of all internet activity in the year 2021, rising from 40.8% in 2020, with 27.7% of all internet traffic being done by the "bad bots" as opposed to the "good bots" that do necessary operations such as web page indexing and automated responses on websites. On the other hand, bad bots are responsible of activities such as web scraping, data mining, spam, data harvesting and other fraudulent operations, for example lead falsification.

In this paper, we tackle the aforementioned issue by proposing the adoption of Machine Learning to recognize bad traffic that passes through a web page by analyzing traffic data gathered through social media advertising and, therefore, measure the real effectiveness of a digital marketing campaign. As such, we propose the classification of the web traffic of online advertising campaigns into "good" or "bad" traffic. This is a different approach with respect to the models previously presented in the related literature. In fact, the models already available in related works focus on analyzing aggregated traffic data instead of classifying specific events, as proposed in this paper. To this end, two models are developed and compared for the classification of the digital advertising traffic i.e., a Shallow

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Neural Network and a Random Forest. By testing different hyperparameters, we compared the performance of the two models over 32,000 samples of real traffic data gathered on various websites from Revelop Srl, an Italian digital marketing company. Specifically, the dataset includes 32,000 rows of data (each row is a sample), evenly split and manually annotated in “good” clicks and “bad” clicks.

The rest of the paper is organized as follows. Section II presents the start of the art about the classification of network traffic for digital marketing and compare already existing studies to the one proposed in this paper. Section III describes the methods used in the presented research, including the proposed classification models as well as the dataset built to train such models and run comparative tests. Section IV gives the details about the experimental evaluation and the computed comparative metrics, discussing the main findings. Finally, Section V concludes the paper and suggests possible future works.

II. Related works

Different studies in scientific literature are related to the analysis of web traffic and the extraction of information from it. For example, Kim and Cho [9] propose a C-LSTM neural network to model both the spacial and temporal information contained in traffic data by combining a Convolutional Neural Network (CNN), a Long Short-Term Memory (LSTM) and a Deep Neural Network (DNN). Combining these three types of neural networks allows detecting anomalies in traffic data during a specific time window, achieving an overall accuracy of 98.6% and a recall of 86.7% on Webscope S5 dataset [10]. Similarly, in response to the increased need for privacy and protection concerning data that passes through the web, especially when it contains sensitive information such as bank account credentials or similar info, Yang et al [11] developed a DNN-based model to classify encrypted traffic, achieving a 97.9% accuracy on a custom dataset built for the presented research. Zhuo et al [12] proposed a similar work: they combine a LSTM and a DNN, adding an autocorrelation coefficient to improve the accuracy, to predict network traffic. They tested their model on three custom datasets built for this purpose, achieving a minimum mean absolute percentage error equal to 1.39%.

Other possible methodologies were investigated by Ma et al [13], which treats URLs as natural language which then gets transformed into mathematical vectors for the model by using statistical laws and natural language processing techniques, and by Jacob et al [14], which uses a Diffusion Convolutional Recurrent Neural Network to model and detect cyber security attacks.

However, the described studies focus on predicting or extracting features from aggregated traffic data. Differently, our study propose to classify singular events on specific websites, instead of relying on aggregated data. To this end, we compare two classifiers, one Shallow Neural Network and a Random Forest, to serve this purpose. Specifically, our paper focuses on:

- Building a new dataset that represents events that occurred on a website, classified into “good” and “bad” clicks.
- Developing data pre-processing functions, to represent the collected data into features processable by the classifiers.
- Developing two classifiers and testing their behaviour on the proposed dataset.

III. Materials and methods

In this work, we propose and compare two different classifiers to recognize single web traffic events on specific websites. An event is defined as a click action that leads on a landing page and records various information about the device that opened the page, such as the IP address, the browser, and the operating system of the device. In this regard, Subsection III-A describes the classifiers proposed in this paper, Subsection III-B presents the data collected for our study, and Subsection III-C details the data pre-processing necessary for our tests.

A. Proposed models

We propose and test the performance of two models for the classification of web traffic events: a Shallow Neural Network and a Random Forest Classifier. The Shallow Neural Network classifier is composed by the following layers:

- A Dense layer with 128 units, using the activation function ‘ReLU’.
- A Dropout layer with a rate of 0.2 to prevent overfitting.
- A Dense layer with 32 units and the same activation function of ‘ReLU’.
- A final Dense layer with 1 unit and ‘sigmoid’ activation function for the binary classification.

The neural network uses Adam as the optimizer during the training. In the experimental evaluation, 0.0001 and 0.0000001 are tested as the learning rate parameter. In addition, an early stopping strategy is implemented by monitoring the minimum validation loss (the binary cross-entropy) testing 3 and 7 as the patience values, in order to avoid overfitting.

Concerning the random forest we develop and test different versions, using a maximum of 500 estimators and a maximum tree depth of 50.

B. Collected dataset

The dataset built for the tests is a collection of real traffic data about clicks gathered on various websites by Revelop Srl, an Italian digital marketing company. Specifically, the dataset includes 32,000 rows of data (each row is a sample, i.e., a web traffic event), evenly split and manually annotated into “good” (label 0) clicks and “bad” (label 1) clicks. The features of each sample are:

- **IP.** The IP address of the device that generated the click.
- **User agent.** The User Agent of the device that generated the click. The User Agent includes the used browser and its version, the device type (desktop or mobile) and the operating system version.

- **Country code.** The ISO code for the country in which the click was recorded.
- **Connection speed.** A string that identifies the type of internet connection of the device. Possible values include broadband connection, mobile connection, and satellite.
- **Access control decision.** A string that identifies how the click to the page was handled by the website: ignored, flagged as suspicious, or accepted.
- **Access control reason code.** The accompanying code that identifies the reason for the access control decision. Possible values include "Flagged as suspicious" or "Non human script".
- **Currency.** If present, a string that identifies the currency used by the transaction click.
- **Device brand.** The brand of the device that generated the click.
- **Device os.** The operating system of the device that generated the click.
- **Datetime.** The timestamp when the click was registered.
- **Pixel refer.** A string that identifies the type of pixel refer used to track the click.
- **Refer.** The referral link that registered the click, if present. Its purpose is to show from where the click came from, saving the entire URL from where the click arrived on the current page.

Table I includes an example of value for each feature in a sample.

TABLE I
EXAMPLE OF FEATURE VALUES FOR AN EVENT OF THE COLLECTED DATASET.

Feature	Value
IP	84.78.240.27
User Agent	Mozilla5.0 (Linux; Android 12; ...)
Country Code	ES
Require connection speed	mobile
Access control decision	accepted
Access control reason code	Non human script
Currency	EUR
Device brand	Samsung
Device os	Android
Datetime	2022-10-24 11:23:15
Pixel refer	Other
Refer	http://somewebsite.net/

C. Data pre-processing

Most of the features composing an event are text strings and, therefore, need some pre-processing in order to be used as numeric input for the classifiers (as training, validation or test data). Specifically:

- The IP is split into the different parts that identify each IP address, the one that identifies the network and the one that identifies the host. Only the first half of the network part was kept and treated as an integer, given that it identifies specific zones. In this way, one can evaluate the presence of a correlation between fraudulent activities and specific zones. On the other hand, the entire IP address is not usable as a feature given that it is not going to be repeated in time.
- In order to avoid a huge number of features by applying one hot encoding on each feature (for each possible value) in the dataset, all the User Agent values are split into different groups according to the device and the version of the used operating system. Each User Agent is labeled as using either a "Current" or an "Old" versions of a device. In facts, a possible sign of fraudulent activity is the adoption of outdated versions of browsers used to run bots and other automated activities. To identify the "current" version of each device, a function reads all the possible values of the user agent in the dataset for a specific device and browser, sorts them according to the version number, drops all the duplicates and assumes that the last value used is the most recent version. Such version is labeled as the "Current" version and all the others are treated as the "Old" version. Thus, a User Agent, which would be normally written as

```
Mozilla/5.0 (Linux; Android 12;
SM-A125F Build/SP1A.210812.016; Wv)
AppleWebKit/537.36 (KHTML, Like Gecko)
Version/4.0 Chrome/105.0.5195.79
Mobile Safari/537.36
[FB_IAB/FB4A;FBAV/386.0.0.35.108;]
```

  simply becomes

```
Mozilla/Linux/Current Version.
```

  Hence, each User Agent in the dataset is represented by a tuple defined by browser/operating system/version.
- From the datetime timestamp, we split the string into its components i.e., day, month, year, hours, minutes, and seconds. Then, we take just the hours, minutes, and seconds, treating them all as integers. This was done to see if the classifiers could find specific patterns in the time of the day where bad traffic could occur. Therefore, the datetime feature is substituted with the 'datetime_hh','datetime_mm' and 'datetime_ss' features. Morevoer, all their values are converted to float32 type and normalized between 0 and 1.
- All the other features are one-hot-encoded, using the dedicated function in the Scikit-learn library.

IV. EXPERIMENTAL EVALUATION, RESULTS, AND DISCUSSION

Labeling one class as positive (for example, the bad events) and the other as negative (for example, the good events), we computed the following metrics for the experimental evaluation of the proposed models:

- Precision i.e., the portion of positive that are correctly identified over all the detected positives.
- Recall i.e., the portion of positives that are correctly identified (over all the available positives).
- Accuracy i.e., the portion of samples that are correctly identified (over all the available samples).

- F_1 score i.e., the armonic mean of precision and recall.

These metrics can be formulated in terms of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) according to the following equations:

$$precision = \frac{TP}{TP + FP} \tag{1}$$

$$recall = \frac{TP}{TP + FN} \tag{2}$$

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

$$F_1\ score = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \tag{4}$$

The tests ran on a PC equipped with an i5-4300U CPU @ 1.90GHz, 16 GB of RAM, without using a GPU, using Keras 2.11.0, TensorFlow 2.11.0, and Scikit-learn 1.2.1.

Different combination of hyperparameters were tried for both models, in order to gather more results and evaluate their influence on the test accuracy. To this end, the collected dataset was randomly spit into training data (90% of all the available data), and test data (10%).

The Shallow Neural Network classifier was trained setting for a maximum of 100 epochs. Four different versions with different hyperparameters were compared. Different learning rates, validation split (which means that a percentage of the training data, randomly selected, was used as the validation set),and early stopping strategy (meaning that the training stopped after a certain number of epochs without any improvement on the minimal validation loss) were selected. We evaluated each version of the neural network on the test dataset, using the Scikit-learn functions to calculate the accuracy metrics and the confusion matrices of the predicted data. To this end, Figures 1 and 2 show the training and validation accuracy and loss during the training. Intuitively, using only 10% of training data for validation i.e., 2,880 samples and the remaining 25,920 samples for training, brings the NN1 and NN3 to a better training, making the neural network solving the classification task. Instead, NN4, where half of the data are for validation and the patience is 7, struggles to converge, and early stopping never occurs.

TABLE II
COMBINATION OF HYPERPARAMETERS WERE USED TO DEVELOP, TRAIN AND COMPARE FOUR DIFFERENT VERSIONS OF THE NEURAL NETWORK

Neural network	Learning rate	Validation split	Patience
NN1	0.0001	0.1	3
NN2	0.0001	0.5	3
NN3	0.0000001	0.1	7
NN4	0.0000001	0.5	7

Concerning the results on the test set, Table III reports the precision, recall, and F_1 score of the four versions of the neural network, whereas Figure 3 shows the confusion matrices.

The NN1 classifier exhibits the best performance, being capable to generalize the classification of both classes, whereas the others performed significantly worse. In facts, NN1 had the best performance both in the reduction of loss and increased accuracy during training and, as showed by the confusion matrix, in predicting the correct value of events. 3714 events were correctly predicted as good traffic and 3727 as bad traffic, whereas only 94 good events were misclassified as bad, and 81 bad events were misclassified as good. Instead, models NN2, NN3 and NN4 show correct prediction of the bad events, but a similarly high number of misclassified good events. Such classifiers overfit on one class and are heavily skewed towards answering with one of the two labels compared to the other.

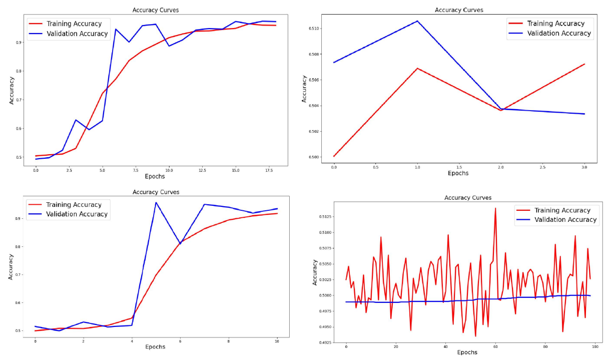

Fig. 1. Training and validation accuracy of the four versions of the Shallow Neural Network classifier, namely NN1 (top-left), NN2 (top-right), NN3 (bottom-left), and NN4 (bottom-right).

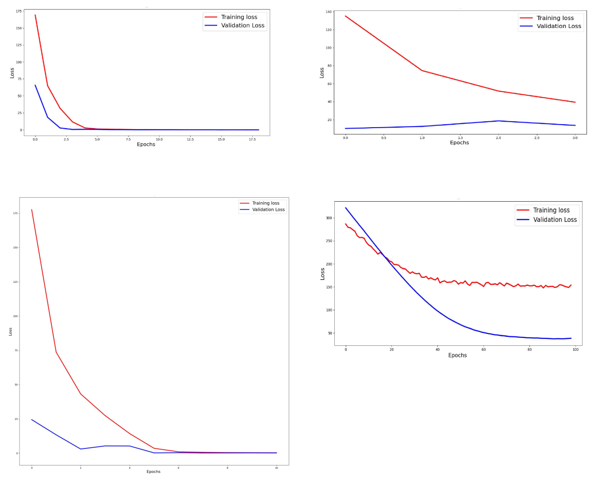

Fig. 2. Training and validation loss of the four versions of the Shallow Neural Network classifier, namely NN1 (top-left), NN2 (top-right), NN3 (bottom-left), and NN4 (bottom-right).

Similarly to the Neural Network classifier, we tested four different versions of the Random Forest, according to the following combinations of parameters:

- 500 Estimators, max depth of the decision trees of 50. We called this classifier RF1.
- 100 Estimators, max depth 5. We called this classifier

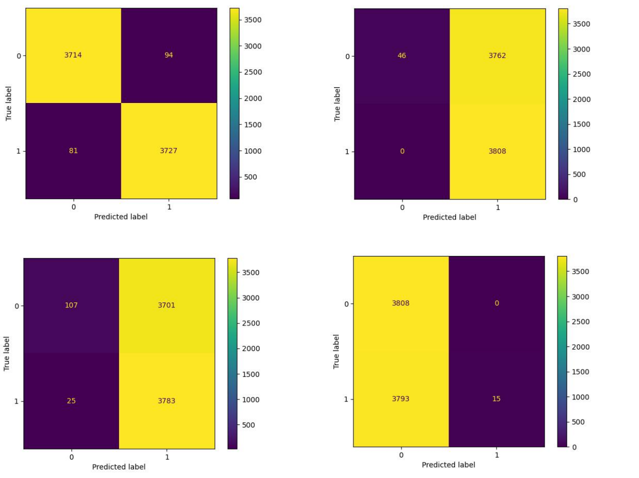

Fig. 3. Confusion matrices about the test set classification for the four versions of the Shallow Neural Network classifier. NN1: top-left. NN2: top-right. NN3: bottom-left. NN4: bottom-right.

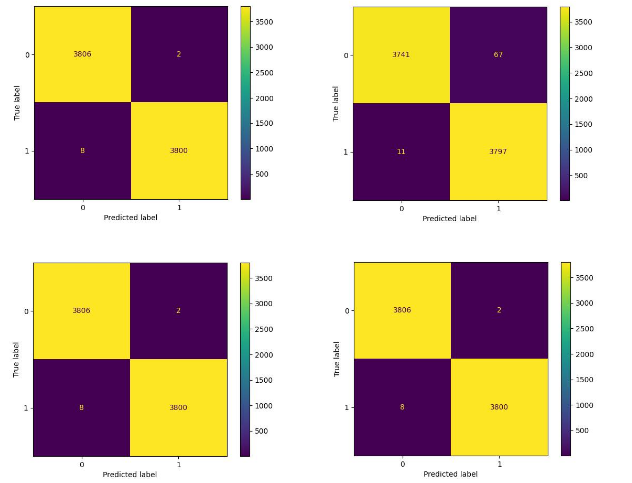

Fig. 4. Confusion matrices about the test set classification for the four versions of the Random Forest classifier. RF1: top-left. RF2: top-right. RF3: bottom-left. RF4: bottom-right.

RF2.

- 250 Estimators, max depth 25. We called this classifier RF3.
- 250 Estimators, max depth 50. We called this classifier RF4.

The dataset splits into training set (90% of the data) and test (10% of the data) were the same used for the Neural Network classifiers. In this regard, Table III includes the computed accuracy metrics for each version of the Random Forest and Figure 4 depicts the confusion matrices about the classification of the samples of the test sets.

TABLE III
PRECISION, RECALL, AND F_1 SCORE ON THE TEST SET FOR THE FOUR VERSIONS OF THE NEURAL NETWORK CLASSIFIER, FOR BOTH CLASSES AND BOTH MODELS

Neural network : Positive class	Precision	Recall	F_1 score
NN1 : “Bad”	0.98	0.98	0.98
NN1 : “Good”	0.98	0.98	0.98
NN2 : “Bad”	1.00	0.01	0.02
NN2 : “Good”	0.50	1.00	0.67
NN3 : “Bad”	0.98	0.98	0.98
NN3 : “Good”	0.98	0.98	0.00
NN4 : “Bad”	0.50	1.00	0.67
NN4 : “Good”	1.00	0.00	0.01
RF1 : “Bad”	1.00	1.00	1.00
RF1 : “Good”	1.00	1.00	1.00
RF2 : “Bad”	1.00	0.98	0.99
RF2 : “Good”	0.98	1.00	0.99
RF3 : “Bad”	1.00	1.00	1.00
RF3 : “Good”	1.00	1.00	1.00
RF4 : “Bad”	1.00	1.00	1.00
RF4 : “Good”	1.00	1.00	1.00

Differently from the Neural Network classifiers, all the Random Forest classifiers achieved uniform results, with high precision and recall. All the confusion matrices show very similar results even with different hyperparameters, which implies that the different numbers of decision trees and max depth do not play a big role in the decision making of all the models. Therefore, such results imply that the classifiers reach their decision in a low number of steps, and the classifier with the lowest number of estimators and max depth (RF2) is more than enough for the proposed classification problem. In facts, most of the samples are classified based on the Refer feature i.e., the referral link that registered the click.

In general, even if the Random Forest classifiers tend to overfit on the collected data, the classification of web traffic events of specific websites into bad or good events is doable, as both classifiers, in different configurations, solve the aforementioned classification problem. Nevertheless, the proposed research suffers of some limitations. Specifically, concerning the Random Forest, more detailed tests with fewer number of estimators should be carried out, to understand if the number of trees can be reduced without compromising the accuracy. Moreover, different classifiers, such as Gradient Boosted Trees, can be tested to further expand the comparison.

V. CONCLUSIONS

In this paper, we proposed a classification of web traffic events of digital market campaigns using Machine Learning. Specifically, we presented a Shallow Neural Network classifier and a Random Forest classifier for such task, running comparative tests for the experimental evaluation on 8 variants of such models, based on the combination of different hyper-parameters. Whereas the Random Forest classifiers get better and more stable accuracy performance with all the veriants, they might overfit, and, thus, they are worth of further studies. Instead, one of the tested neural networks converge to similar performance, but the loss curve does not show overfitting.

As such, the proposed classification of web traffic events of specific marketing campaigns is feasible.

Future works will tackle the limitations identified in this study. Specifically, further tests with the Random Forest classifiers, as well as with other classifiers, and a feature analysis would allow getting an in-depth view of the needed data, in effort to go toward a more efficient classification, without losing accuracy.

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